3 **(a)** **(i)** force is rate of change of momentum .... B1 [1]

(ii) force on body A is equal in magnitude to force on body B (from A) M1
forces are in opposite directions A1
forces are of the same kind ....A1 [3]

(b) **(i)** 1 $F_A = -F_B$ .... B1 [1]
2 $t_A = t_B$ .... B1 [1]

(ii) $\Delta p = F_A t_A = -F_B t_B$ .... B1 [1]

(c) graph: momentum change occurs at same times for both spheres B1
final momentum of sphere B is to the right M1
and of magnitude 5 N s .... A1 [3]

4 **(a)** e.g. no energy transfer